# Geometric Strategies for Sharp-Interface Front-Tracking Methods

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**Abstract**

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## 1 Notes

- Pure front-tracking, no coupling to a solver. - Bibliography, litterature review of current Front tracking method, markers maintenance, hygiene, - The differents front tracking problems (merging, breakup, redistribution, insertion, ....) - Algorithms - why only normal component ? Because fundamentally, front is moving with front velocity, which is normal to the front. - Only normal component of velocity, no tangential redistribution =¿ What's going one ? Add redistribution =¿ better ? - Numerical Tests - DEC, DDG - References

## 2 Introduction

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## 3 Continuum Sharp interface formulation

Let $\Omega \subset \mathbb{R}^d$ be a fixed domain, with $d \in \{2, 3\}$, and let $\Gamma(t) \subset \Omega$ be a moving interface separating two distinct phases. The interface $\Gamma(t)$ is a codimension-one manifold, that is, a curve when $d = 2$ and a surface when $d = 3$. The interface evolves according to a normal velocity field $V_n$ defined on $\Gamma(t)$, which may depend on the local geometry of the front and on physical fields such

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as pressure, temperature, or concentration. The governing equation for the interface motion is therefore

$$\frac{\partial \Gamma}{\partial t} = V_n n, \tag{1}$$

where $n$ is the unit normal vector field on $\Gamma(t)$ pointing from one phase to the other. $V_n$ may include contributions from curvature, external forces, or other physical effects, depending on the specific problem under consideration.

# 4 Lagrangian front-tracking representation and intrinsic geometric operators

In the present work, the moving sharp interface is represented explicitly by a Lagrangian front mesh. The front is the primary geometric unknown: all normals, curvatures, intrinsic differential operators, are constructed from this mesh. The purpose of this section is therefore twofold: (i) to define the minimal discrete data required to represent the interface in two and three space dimensions, and (ii) to introduce the intrinsic geometric and differential operators that can be constructed directly on this front.

The front-tracking representation described here concerns codimension-one interfaces in $d \in \{2, 3\}$, namely polygonal curves in two dimensions and triangulated surfaces in three dimensions.

At time $t$, the discrete interface is denoted by $\Gamma_h(t)$, and its vertex positions are gathered in the vector

$$X(t) = \{x_a(t)\}_{a=1}^{N_v}. \tag{2}$$

Here $h$ denotes the characteristic front-mesh size, and $N_v$ is the number of front vertices. The notation emphasizes that $\Gamma_h(t)$ is a discrete approximation of the smooth interface $\Gamma(t)$ introduced in Section 3. For simplicity, the explicit dependence on time is omitted below whenever no ambiguity arises.

## 4.1 Discrete front mesh

The discrete front representation depends on the ambient space dimension.

**Two-dimensional front.** When $d = 2$, the interface is represented by a polygonal curve

$$\Gamma_h = (V_h, E_h), \tag{3}$$

where $V_h = \{x_a\}_{a=1}^{N_v} \subset \mathbb{R}^2$ is the set of front vertices and $E_h = \{e\}_{e=1}^{N_e}$ is the set of oriented edges. Each edge $e = (i, j)$ connects two consecutive vertices and is oriented from $x_i$ to $x_j$.

**Three-dimensional front.** When $d = 3$, the interface is represented by a triangulated surface

$$\Gamma_h = (V_h, F_h), \tag{4}$$

where $V_h = \{x_a\}_{a=1}^{N_v} \subset \mathbb{R}^3$ is the set of front vertices and $F_h = \{f\}_{f=1}^{N_f}$ is the set of oriented triangular faces. Each face $f = (i, j, k)$ is oriented consistently with the global orientation of the surface.

In both cases, the front mesh is fully determined by the vertex coordinates $x_a$, the mesh connectivity and a consistent orientation. No additional geometric unknown is introduced at this stage. All normals, curvatures, and intrinsic operators are reconstructed from these quantities.

Unless stated otherwise, $\Gamma_h$ is assumed to be an orientable manifold mesh, closed for embedded internal interfaces and possibly with boundary for open-front test cases. In particular, the mesh connectivity is assumed to be compatible with a globally consistent orientation.

## 4.2 Topology, orientation, and adjacency

A front mesh is not merely a collection of points embedded in physical space: it also carries a combinatorial structure that determines local neighborhoods, orientation signs, and discrete differential stencils.

For a curve mesh, the relevant topological information is the vertex-edge incidence. For a triangulated surface, one additionally needs:

- the set of unique edges,
- the vertex-edge adjacency,
- the vertex-face adjacency,
- the edge-face adjacency,
- the orientation of each face relative to its boundary edges.

This structure is encoded by incidence operators. For both curves and surfaces, the discrete exterior derivative from vertices to edges is

$$d_0 : C^0(\Gamma_h) \longrightarrow C^1(\Gamma_h), \tag{5}$$

where $C^0(\Gamma_h)$ denotes vertex-based scalar fields and $C^1(\Gamma_h)$ denotes oriented edge-based quantities. For a scalar field $u_h = \{u_a\}_{a=1}^{N_v}$,

$$(d_0 u_h)_e = u_j - u_i, \qquad e = (i, j). \tag{6}$$

In three dimensions, one further introduces the discrete exterior derivative from edges to faces,

$$d_1 : C^1(\Gamma_h) \longrightarrow C^2(\Gamma_h), \tag{7}$$

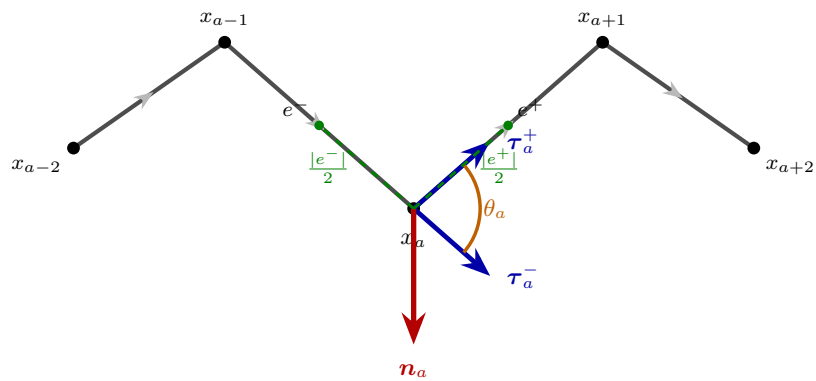
where $C^2(\Gamma_h)$ denotes face-based quantities. For a consistently oriented triangulated surface, the identity

$$d_1 d_0 = 0 \tag{8}$$

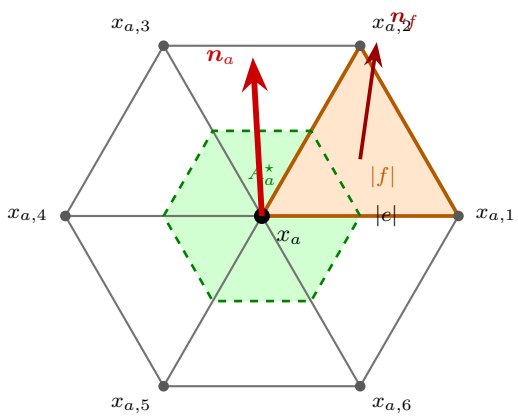
holds exactly at the discrete level.

This topological layer is essential for the construction of intrinsic geometric operators on the front. It also provides natural consistency checks, such as closedness, manifoldness, and global orientation coherence.

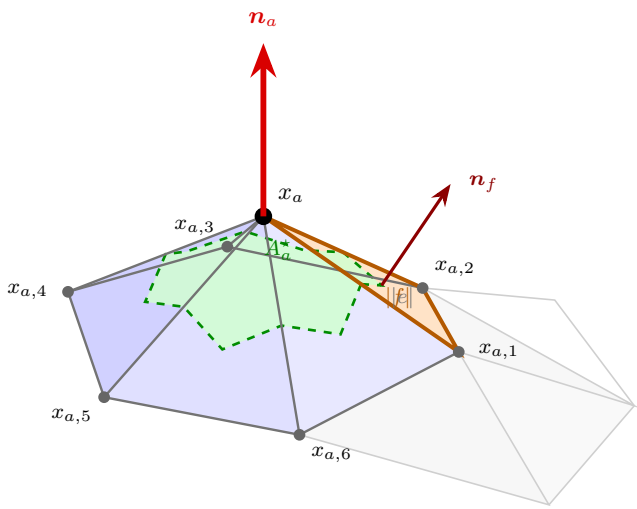
## 4.3 Minimal geometric data




$$\ell_a^\star = \frac{|e^-| + |e^+|}{2}, \qquad \kappa_a = \frac{\theta_a}{\ell_a^\star}$$




$$\boldsymbol{n}_a = \frac{\sum_{f \ni a} |f|\, \boldsymbol{n}_f}{\left\| \sum_{f \ni a} |f|\, \boldsymbol{n}_f \right\|} \qquad A_a^\star = \sum_{f \ni a} \frac{|f|}{3}$$




$$\boldsymbol{n}_a = \frac{\sum_{f \ni a} |f|\, \boldsymbol{n}_f}{\left\| \sum_{f \ni a} |f|\, \boldsymbol{n}_f \right\|} \qquad A_a^\star = \sum_{f \ni a} \frac{|f|}{3}$$

Starting from the embedded front mesh, one reconstructs the minimal set of metric quantities needed to define intrinsic operators.

### 4.3.1 Polygonal curves in two dimensions

Let $e = (i, j) \in E_h$ be an oriented edge of the polygonal curve. Its length and unit tangent are

$$|e| = \|x_j - x_i\|, \qquad \tau_e = \frac{x_j - x_i}{|e|}. \tag{9}$$

At each vertex $x_a$, we define the dual vertex length

$$\ell_a^\star = \frac{1}{2} \sum_{e \ni a} |e|, \tag{10}$$

that is, half the sum of the lengths of the edges incident to that vertex. This quantity plays the role of a vertex-based measure along the discrete curve.

The minimal metric data carried by a polygonal front are therefore

$$\mathcal{G}_h^{\text{curve}} = \{|e|,\ \tau_e,\ \ell_a^\star\}\,. \tag{11}$$

### 4.3.2 Triangulated surfaces in three dimensions

Let $f = (i, j, k) \in F_h$ be an oriented triangular face. Its non-normalized normal vector is

$$\widetilde{n}_f = (x_j - x_i) \times (x_k - x_i), \tag{12}$$

and its unit normal and area are

$$n_f = \frac{\widetilde{n}_f}{\|\widetilde{n}_f\|}, \qquad |f| = \frac{1}{2}\|\widetilde{n}_f\|. \tag{13}$$

For each edge $e = (i, j)$, the edge length is

$$|e| = \|x_j - x_i\|. \tag{14}$$

At each vertex $x_a$, one defines a dual vertex area $A_a^\star$. A simple choice is the barycentric dual area

$$A_a^\star = \sum_{f \ni a} \frac{|f|}{3}, \tag{15}$$

obtained by assigning one third of each incident triangle area to each of its vertices. More elaborate mixed dual areas may also be used, but the present notation remains unchanged.

The minimal metric data carried by a triangulated front are therefore

$$\mathcal{G}_h^{\mathrm{surf}} = \{|e|,\ |f|,\ A_a^\star,\ n_f,\ \omega_e\}, \tag{16}$$

where $\omega_e$ denotes the edge-based metric coefficient entering the discrete Hodge star on tangential 1-forms. For the standard cotangent Laplace–Beltrami operator on a triangulated surface, $\omega_e$ is the usual cotangent weight associated with the edge $e$.

## 4.4 Dual measures and Hodge stars

The incidence operators defined above are purely combinatorial. Metric information enters through discrete Hodge stars, which weight primal cochains by the appropriate dual measures.

For a polygonal curve, the vertex Hodge star is

$$\star_0 = \mathrm{diag}\big(\ell_1^\star, \ldots, \ell_{N_v}^\star\big), \tag{17}$$

while the edge Hodge star is

$$\star_1 = \mathrm{diag}\big(|e_1|^{-1}, \ldots, |e_{N_e}|^{-1}\big). \tag{18}$$

For a triangulated surface, the vertex Hodge star is

$$\star_0 = \mathrm{diag}\big(A_1^\star, \ldots, A_{N_v}^\star\big), \tag{19}$$

and the edge Hodge star is

$$\star_1 = \mathrm{diag}\big(\omega_{e_1}, \ldots, \omega_{e_{N_e}}\big), \tag{20}$$

where $\omega_e$ denotes the edge-based metric coefficient associated with the discrete tangential 1-form on edge $e$. In the standard cotangent formulation, $\omega_e$ is the usual cotangent weight.

If face-based 2-cochains are needed, the face Hodge star is

$$\star_2 = \mathrm{diag}\big(|f_1|, \ldots, |f_{N_f}|\big). \tag{21}$$

These Hodge stars provide the metric structure required to define intrinsic differential operators on the front.

## 4.5 Intrinsic differential operators on the front

### 4.5.1 Tangential projection

Let $n$ denote a unit normal field on the front. For any ambient vector $v \in \mathbb{R}^d$, its tangential projection onto the local tangent space of the interface is

$$\Pi_\Gamma(v) = v - (v \cdot n)n. \tag{22}$$

This decomposition separates normal motion of the interface from tangential redistribution along the front.

### 4.5.2 Laplace-Beltrami operator

Let $u_h \in C^0(\Gamma_h)$ be a scalar field defined at the front vertices. Its intrinsic Laplace-Beltrami operator is written in discrete exterior calculus form as

$$\Delta_{\Gamma_h} u_h = \star_0^{-1} d_0^\top \star_1 d_0 u_h. \tag{23}$$

This formula applies both to polygonal curves and triangulated surfaces, provided the edge Hodge star $\star_1$ is chosen consistently with the underlying metric. For a curve, (23) approximates the second derivative along arclength. For a triangulated surface, it yields the standard edge-weighted discrete surface Laplacian, including the cotangent form as a particular case. The operator is intrinsic in the sense that it depends only on the metric and topology of the front mesh.

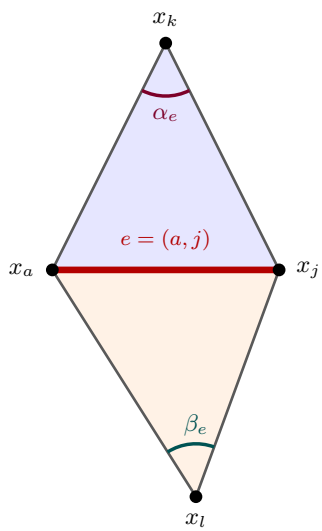
### 4.5.3 Surface gradient on a triangulated surface

When $d = 3$, a scalar field $u_h = \{u_a\}_{a=1}^{N_v}$ defined at the vertices of a triangulated surface induces a piecewise constant tangential gradient on each face. For a face $f = (a, b, c)$, one has

$$\nabla_{\Gamma_h} u_h\big|_f = \frac{1}{2|f|}\left[u_a\, n_f \times (x_c - x_b) + u_b\, n_f \times (x_a - x_c) + u_c\, n_f \times (x_b - x_a)\right]. \tag{24}$$

This expression is exact for piecewise affine scalar fields on the triangulation.

Together, the tangential projection, the Laplace-Beltrami operator, and the surface gradient provide the core intrinsic differential tools needed later for the front evolution equations.




$$\omega_e = \frac{\cot \alpha_e + \cot \beta_e}{2}$$

$$(\Delta_{\Gamma_h} u_h)_a = \frac{1}{A_a^\star} \sum_{j \sim a} \omega_{(a,j)} \, (u_j - u_a) \qquad \text{(DEC form: } \star_0^{-1} d_0^\top \star_1 \, d_0 \, u_h)$$

## 4.6 Normals and curvature

We now introduce the geometric quantities derived from the front representation.

### 4.6.1 Normals and curvature of a polygonal curve

Let $x_a$ be a vertex of the polygonal curve, and let $\tau_a^-$ and $\tau_a^+$ denote the incoming and outgoing unit tangents of the incident edges. A discrete vertex tangent is obtained from their normalized average, and the corresponding unit normal $n_a$ is obtained by a quarter-turn rotation in the plane. The choice of rotation sign is fixed globally so as to remain consistent with the interface normal convention of Section 3.

The discrete signed curvature at vertex $x_a$ is defined by

$$\kappa_a = \frac{\theta_a}{\ell_a^\star}, \tag{25}$$

where $\theta_a$ is the signed turning angle from $\tau_a^-$ to $\tau_a^+$ and $\ell_a^\star$ is the dual vertex length defined in (10). This quantity converges to the continuous signed curvature as the polygonal front is refined.

### 4.6.2 Normals and curvature of a triangulated surface

For a triangulated surface, the face normals $n_f$ are already defined in (13). A vertex normal is obtained by averaging the normals of the incident faces with

area weights,

$$n_a = \frac{\sum_{f \ni a} |f|\, n_f}{\left\| \sum_{f \ni a} |f|\, n_f \right\|}. \tag{26}$$

The discrete mean-curvature normal is then defined from the Laplace–Beltrami operator applied componentwise to the embedding $X = \{x_a\}_{a=1}^{N_v}$:

$$\boldsymbol{H}_a = -\frac{1}{2} \left(\Delta_{\Gamma_h} X\right)_a . \tag{27}$$

Its scalar projection on the vertex normal gives the discrete mean curvature,

$$H_a = \boldsymbol{H}_a \cdot n_a. \tag{28}$$

This convention is consistent with the continuous identity $\Delta_\Gamma X = -2Hn$. If one uses the twice-mean-curvature convention $\kappa = 2H$, then the corresponding curvature normal is $\boldsymbol{\kappa}_a = 2\boldsymbol{H}_a$.

The Gaussian curvature is defined through the angle defect formula

$$K_a = \frac{1}{A_a^\star} \left( 2\pi - \sum_{f \ni a} \theta_a^f \right), \tag{29}$$

where $\theta_a^f$ denotes the interior angle of face $f$ at vertex $x_a$. The quantities $n_a$, $H_a$, and $K_a$ form the minimal curvature data attached to a triangulated surface front.

## 4.7 Global geometric diagnostics

The front representation also provides global geometric quantities that are useful for verification and monitoring.

**Front measure.** For a polygonal curve, the total front length is

$$|\Gamma_h| = \sum_{e \in E_h} |e|, \tag{30}$$

whereas for a triangulated surface, the total front area is

$$|\Gamma_h| = \sum_{f \in F_h} |f|. \tag{31}$$

**Enclosed measure.** If the front is closed, it also defines an enclosed geometric measure: an enclosed area in two dimensions and an enclosed volume in three dimensions.

**Topological diagnostics.** For a closed triangulated surface, the Euler characteristic is

$$\chi = N_v - N_e + N_f, \tag{32}$$

where $N_e$ is the number of unique edges. Combined with the discrete Gaussian curvature, it yields the Gauss-Bonnet relation

$$\sum_{a=1}^{N_v} K_a A_a^\star = 2\pi\chi, \tag{33}$$

for a closed triangulated surface without boundary, up to floating-point roundoff. This identity is a useful diagnostic for curvature consistency and mesh integrity.

The front-based quantities introduced above are intrinsic to the Lagrangian mesh.

# References


[1] Désir-André Koffi Bi, Mathilde Tavares, Eric Chénier, and Stéphane Vincent. Accuracy and convergence of the curvature and normal vector discretizations for 3D static and dynamic front-tracking interfaces. *Journal of Computational Physics*, 461:111197, July 2022.